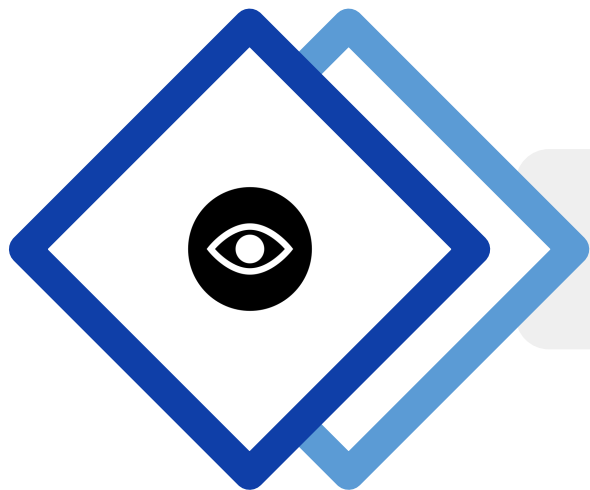


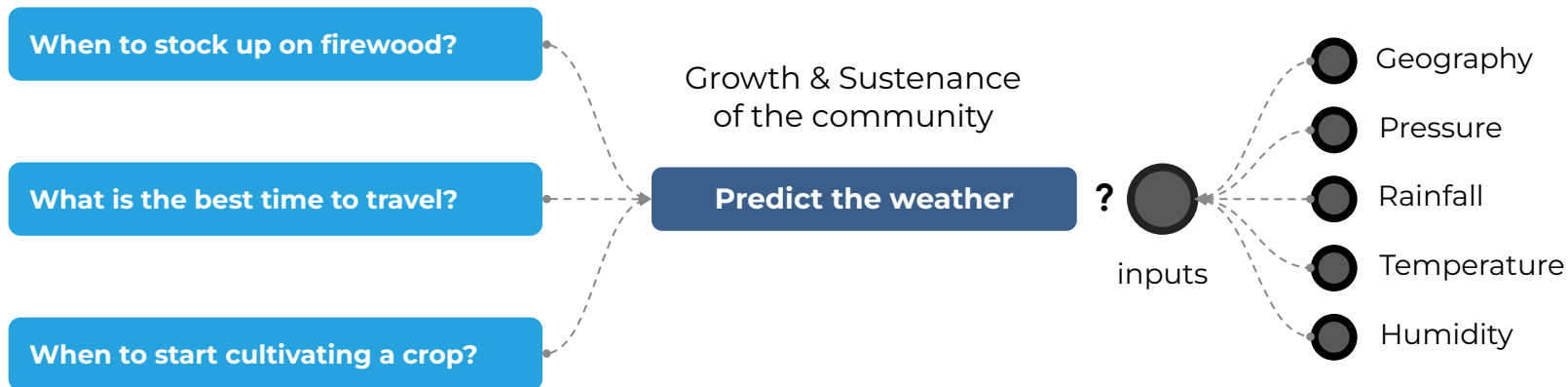


Introduction to the World of Data

Origin of Decisions

Decisions were always **data-driven**

Let's consider few situations that early civilizations might have faced



Decisions are made today by businesses the same way - but the methods have become more **accurate** and **faster** owing to the evolution of **statistical techniques** & **computing capabilities**

Paradigms in data science

Inferential

Make predictions on population based on sample data

Use statistical methods to draw conclusions / infer from data

Representativeness of data

1. Effectiveness of a new medication through randomized trial
2. Impact of a new policy on citizens

Computational

Leverage computational methods and technology to scale insight generation

Implement algorithms and computational methods to analyse data

Complexity of algorithms and cost of training large models

1. Weather forecasting based on historical and weather patterns
2. Optimize routing of vehicles to minimize costs

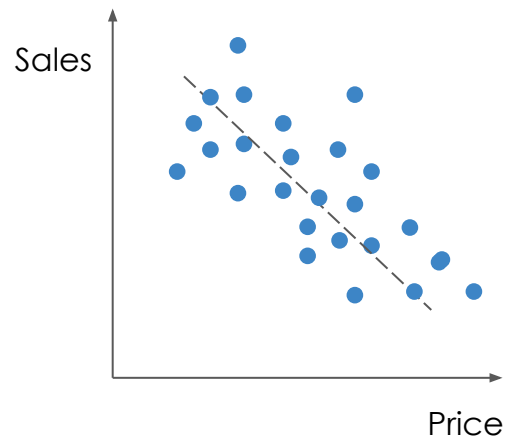
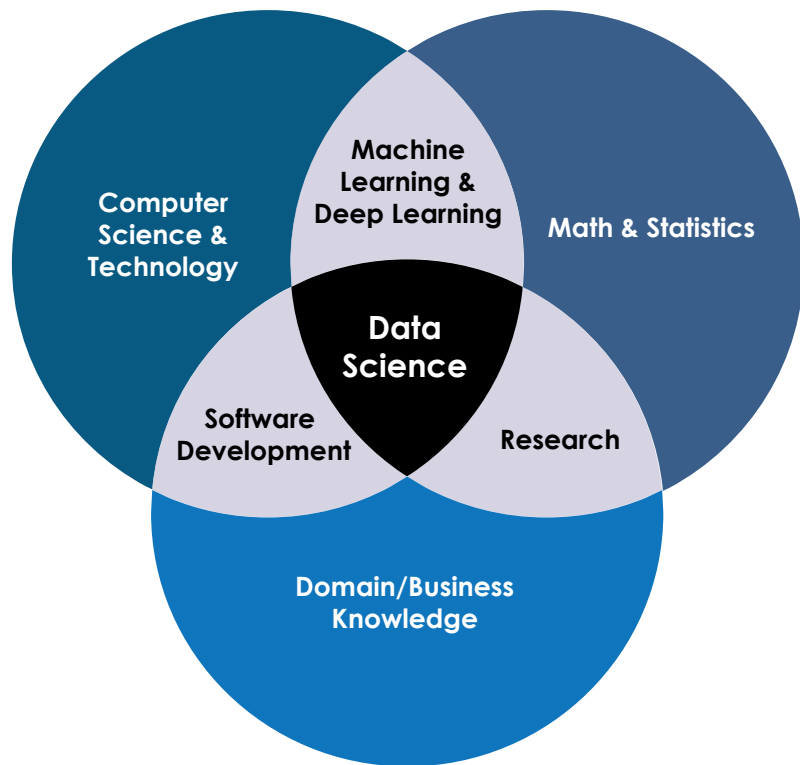
 Focus

 Methods

 Limitations

 Examples

The need for math in Data Science



$$\text{Sales} = 500 - 20 * \text{Price}$$

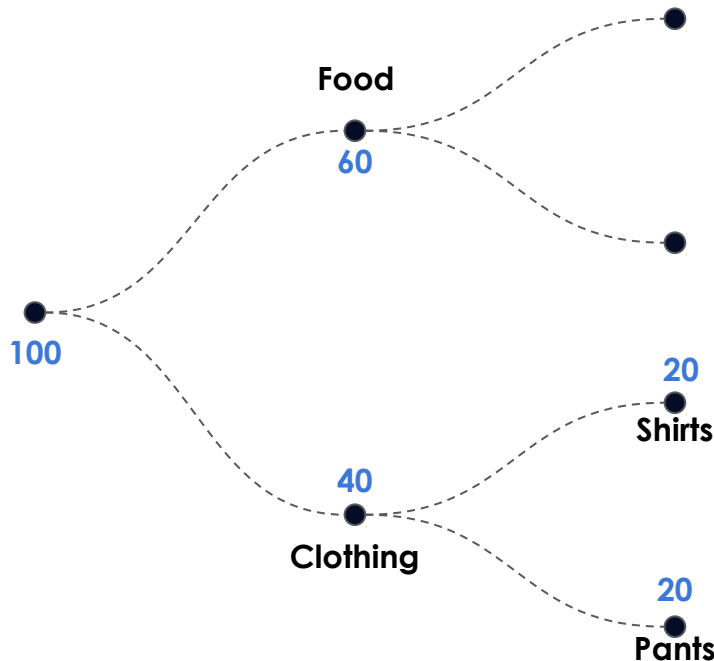


Probability & Descriptive Statistics

Bayes' Rule

“Determine **probability** of a hypothesis based on prior knowledge and new evidence.”

Eg: Shopping patterns of 100 people in a retail store



Probability of a new person to enter the food section?

Probability (Food) = 0.6

Probability of a person to enter the clothing section?

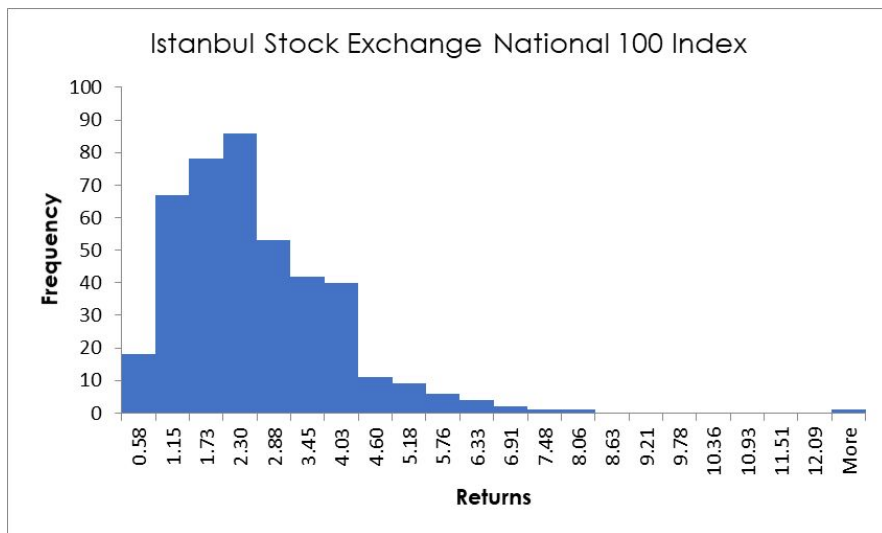
Probability (Clothing) = 0.4

Probability of a person in the shirt section, given, he is in the clothing section?

Probability (Shirt | Clothing) = 0.5

Descriptive Statistics

“Describing features of a dataset by generating summaries about data samples ”



Average Return = Mean

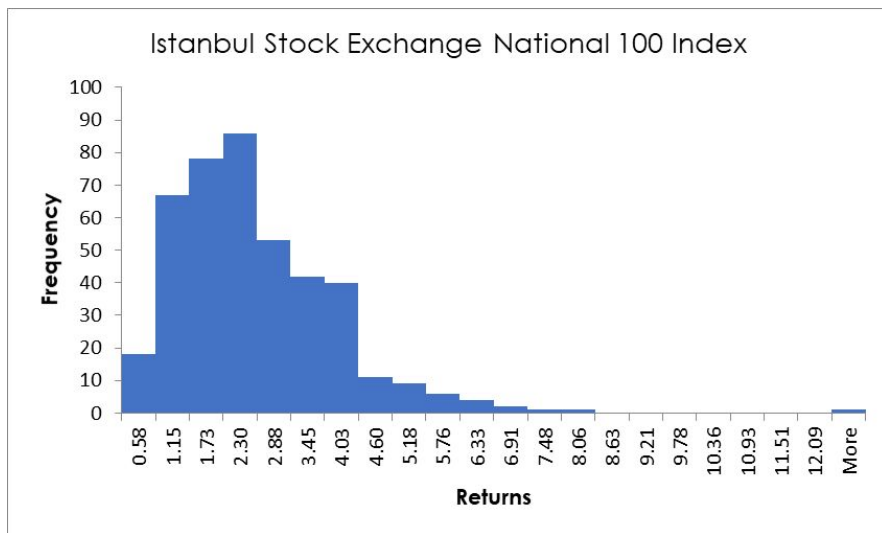
$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Risk = Standard Deviation

$$\sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$$

Descriptive Statistics

“Describing features of a dataset by generating summaries about data samples ”





Happy Learning !

